

Lecture 06: Theory of Generalization

Bounding the Growth Function and the VC Inequality

Caltech CS156: Learning from Data

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1 Overview & Mathematical Motivation

In previous lectures, we established that evaluating whether learning is feasible requires bounding the out-of-sample error E_{out} using the in-sample error E_{in} . However, standard Hoeffding bounds apply only to a single fixed hypothesis. When an algorithm $\mathcal{A}$ selects the best hypothesis g from a candidate set $\mathcal{H}$ of size M , applying the Union Bound yields:

$$P(|E_{\text{in}}(g) - E_{\text{out}}(g)| > \epsilon) \leq 2Me^{-2\epsilon^2 N} \quad (1)$$

When $\mathcal{H}$ is infinite ($M = \infty$), this bound becomes completely vacuous. Lecture 06 addresses this issue by replacing the infinite cardinality M with a finite, polynomially bounded quantity: the **Growth Function** $m_{\mathcal{H}}(N)$.

2 Dichotomies and Break Points

To count hypothesis variations over a finite sample $\mathcal{D} = \{\mathbf{x}_1, \dots, \mathbf{x}_N\}$, we define a **dichotomy** as a hypothesis restricted to $\mathcal{D}$:

$$\mathcal{H}(\mathbf{x}_1, \dots, \mathbf{x}_N) = \{(h(\mathbf{x}_1), \dots, h(\mathbf{x}_N)) \mid h \in \mathcal{H}\} \subseteq \{-1, +1\}^N \quad (2)$$

The **Growth Function** $m_{\mathcal{H}}(N)$ is the maximum number of dichotomies that $\mathcal{H}$ can generate on any N points:

$$m_{\mathcal{H}}(N) = \max_{\mathbf{x}_1, \dots, \mathbf{x}_N \in \mathcal{X}} |\mathcal{H}(\mathbf{x}_1, \dots, \mathbf{x}_N)| \leq 2^N \quad (3)$$

Definition: Break Point

A integer k is called a **break point** for $\mathcal{H}$ if no set of k data points can be shattered by $\mathcal{H}$, meaning $m_{\mathcal{H}}(k) < 2^k$.

For example:

- **Positive Rays:** $m_{\mathcal{H}}(N) = N + 1 \implies$ Break point $k = 2$.
- **Positive Intervals:** $m_{\mathcal{H}}(N) = \frac{1}{2}N^2 + \frac{1}{2}N + 1 \implies$ Break point $k = 3$.
- **2D Perceptron:** $m_{\mathcal{H}}(N)$ shatters 3 points ($2^3 = 8$), but fails to shatter 4 points (XOR pattern) $\implies$ Break point $k = 4$.

3 Bounding the Growth Function: Combinatorial Quantity $B(N, k)$

To prove that $m_{\mathcal{H}}(N)$ grows polynomially if a break point exists, we introduce the purely combinatorial quantity $B(N, k)$: the maximum number of dichotomies on N points such that no subset of k points is shattered.

Bounding Theorem (Sauer-Shelah Lemma)

If a hypothesis set $\mathcal{H}$ has a break point k , then for all N :

$$m_{\mathcal{H}}(N) \leq B(N, k) = \sum_{i=0}^{k-1} \binom{N}{i} \quad (4)$$

3.1 Recursive Induction Proof

We construct a matrix of size $B(N, k) \times N$ with rows representing valid dichotomies. Isolate the last column $\mathbf{x}_N$ and partition the rows into two groups:

- S_1 (α rows): Patterns on $\mathbf{x}_1, \dots, \mathbf{x}_{N-1}$ that appear with only **one** extension (+1 or -1).
- S_2 (2β rows): Patterns on $\mathbf{x}_1, \dots, \mathbf{x}_{N-1}$ that appear with **both** extensions (+1 and -1).

Total rows $B(N, k) = \alpha + 2\beta = (\alpha + \beta) + \beta$.

1. Looking at the truncated matrix on $N - 1$ points, the total unique patterns $\alpha + \beta \leq B(N - 1, k)$.
2. Looking at S_2 alone on $N - 1$ points, if S_2 shattered $k - 1$ columns, then adding $\mathbf{x}_N$ (which has both +1 and -1) would shatter k columns in the full matrix, violating the break point k . Thus $\beta \leq B(N - 1, k - 1)$.

Summing these yields the core recurrence relation:

$$B(N, k) \leq B(N - 1, k) + B(N - 1, k - 1) \quad (5)$$

Solving this recurrence by mathematical induction yields the binomial sum $\sum_{i=0}^{k-1} \binom{N}{i}$, which is a polynomial in N of degree $k - 1$:

$$B(N, k) \leq \frac{N^{k-1}}{(k-1)!} + \mathcal{O}(N^{k-2}) \implies \text{Polynomial Growth!} \quad (6)$$

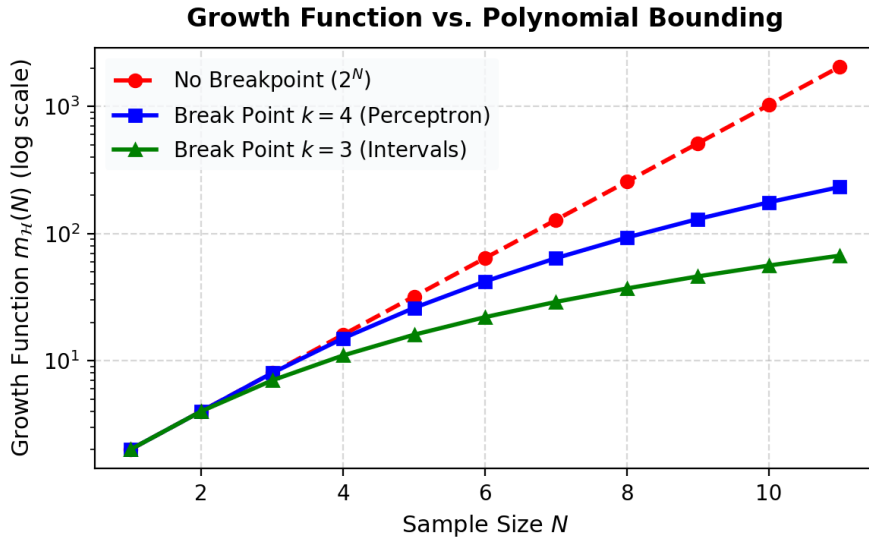


Figure 1: Growth function $m_{\mathcal{H}}(N)$ bounded by polynomial $B(N, k)$ when a break point k exists.

4 The Vapnik-Chervonenkis (VC) Inequality

Replacing M with $m_{\mathcal{H}}(N)$ directly in Hoeffding's inequality is invalid because E_{out} is defined over the continuous input space $\mathcal{X}$, not just N points.

To overcome this, Vapnik and Chervonenkis introduced a second "ghost sample" $\mathcal{D}'$ of size N . By bounding the probability of discrepancy between $E_{\text{in}}(\mathbf{w})$ and $E'_{\text{in}}(\mathbf{w})$, we restrict all evaluations to $2N$ points, yielding the fundamental theoretical result of Machine Learning:

Vapnik-Chervonenkis (VC) Inequality

For any hypothesis set $\mathcal{H}$, target f , distribution $P(\mathbf{x})$, and tolerance $\epsilon > 0$:

$$P \left(\sup_{h \in \mathcal{H}} |E_{\text{in}}(h) - E_{\text{out}}(h)| > \epsilon \right) \leq 4 m_{\mathcal{H}}(2N) e^{-\frac{1}{8} \epsilon^2 N} \quad (7)$$

As long as a break point k exists, $m_{\mathcal{H}}(2N)$ grows polynomially as $\mathcal{O}((2N)^{k-1})$, while the negative exponential $e^{-\frac{1}{8} \epsilon^2 N}$ eventually dominates, ensuring that $E_{\text{in}} \rightarrow E_{\text{out}}$ as $N \rightarrow \infty$.